Math 242, S13 **Name:** Answer

Quiz 4 **SSN:** xxx-xx-

Instructions: **There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (10 points) Find a general solution for the following differential equation using the substitution method:

$$\frac{dy}{dx} = (x+y)^2$$

(*Hint: Use the substitution* $v = x + y$.)

Solution: Let $V = X + y$, then $y = V - X$ and

$$\frac{dy}{dx} = \frac{dV}{dx} - 1$$

Thus we transform the original equation into

$$\frac{dV}{dx} - 1 = V^2$$

$$\frac{dV}{dx} = V^2 + 1$$

$$\frac{1}{V^2+1}\,dV = dx$$

$$\int \frac{1}{V^2+1}\,dV = \int dx$$

$$\arctan V = X + C$$

$$V = \tan(X + C)$$

So we get

$$y(x) = V - X = \underline{\tan(X + C) - X}$$